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Early detection of chronic kidney disease based on a SURD-enhanced machine learning model

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Chronic kidney disease (CKD) represents a major global health burden, and early, reliable risk prediction remains clinically challenging. This study proposes a CKD prediction framework that integrates machine learning with Synergy–Unique–Redundant Decomposition (SURD) from causal information theory to enhance both predictive performance and interpretability. Ten classification models were developed using the UCI-CKD dataset (n = 400). Missing values were handled using multiple imputation via chained equations, and class imbalance was addressed with the synthetic minority oversampling technique. Model performance was evaluated using accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC). To rigorously assess generalizability and mitigate concerns regarding overfitting, extensive external validation was conducted using a large-scale real-world electronic health record cohort from the MIMIC-IV database (n = 27,834). While several models achieved near-perfect performance on the internal dataset, the Random Forest model demonstrated superior generalization in the external cohort, achieving an AUC of 0.990 (95% CI 0.989–0.991), compared with an AUC of AUC: 0.914 (95% CI0.912–0.916) for the baseline Decision Tree. SURD-based causal decomposition and feature importance analyses consistently identified clinically established predictors, including serum creatinine and hemoglobin. Overall, these results indicate that the proposed SURD-guided framework provides a robust and interpretable approach for early CKD risk stratification and demonstrates stable performance when transferred from benchmark datasets to real-world clinical settings.

Keywords Chronic kidney disease (CKD), SURD (synergistic unique redundant decomposition), Causal inference, Machine learning

Chronic kidney disease (CKD) is a progressive condition characterized by high prevalence and mortality[1]. Due to its insidious onset, early-stage CKD often lacks obvious clinical symptoms, leading to delayed diagnoses and eventual progression to end-stage renal disease[2]. While the standard diagnostic criterion involves an estimated glomerular filtration rate (eGFR) below 60 mL/min/1.73 m[23], early identification through routine blood and urine tests remains a more practical screening approach for general populations[4,5].

While machine learning has significantly improved CKD prediction accuracy, model complexity often compromises clinical transparency. Existing XAI tools typically evaluate features in isolation, which fails to account for the redundant or synergistic interactions between clinical variables[6]. Because statistical correlations fail to identify the causal structure of CKD, we utilize a causal framework to resolve the high redundancy and interactions among clinical features.

Traditional causal discovery methods, including constraint-based and hybrid algorithms[7–10], have laid the foundation for identifying directional dependencies. However, established techniques such as Granger Causality (GC)[11], Convergent Cross-Mapping (CCM)[12], and Transfer Entropy[13]—while effective in specific domains—often treat information flow as an aggregate signal. These methods, along with recent conditional independence tests like PCMCI[14], struggle to handle the non-linearities and unobserved confounders typical of physiological data.

To address these limitations, this study utilizes Synergy-Unique-Redundant Decomposition (SURD)[15]. Unlike standard causal frameworks, SURD facilitates a granular decomposition of causal relationships, allowing us to isolate independent predictors from overlapping clinical signals. This approach aligns with the recent

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emphasis on hybrid XAI and data balancing in medical diagnostics[16,17], moving beyond associative patterns to provide a more biologically plausible explanation for model decisions.

XAI provides a framework for interpreting machine learning decisions, addressing the opacity often caused by complex clinical correlations. While standard ML models excel at identifying associations, they lack the capacity to elucidate causal mechanisms. By applying causal inference, this study investigates the underlying causal relationships between CKD markers to complement the findings of traditional associative models. To this end, we investigate the following research questions:

1. Does applying the SURD decomposition results to feature selection improve the generalization ability and diagnostic accuracy of machine learning models?
2. Is SURD able to identify the key physiological features that have the most synergistic or unique effects on the development of CKD?
3. Can XAI technology improve the transparency and clinical usability of CKD prediction models?

Related works

In previous studies, causal inference and machine learning have been increasingly combined to support disease diagnosis, risk prediction, and treatment evaluation. MINDMerge[18] was proposed to address inconsistency across causal discovery algorithms by fusing multiple causal models with credibility weighting and loop-elimination, and was validated on synthetic networks and ICU databases (eICU and MIMIC-III), showing improved causal graph recovery and strong performance in AKI risk prediction. Other studies integrated causal discovery (e.g., LiNGAM) with conventional predictive models[19], or introduced temporal causal reasoning by combining VAR with GNNs to enhance discrimination across standard classifiers[20]. Beyond causal discovery, observational EHR-based designs have been used to emulate trial-like inference: regression discontinuity coupled with dynamic Bayesian networks revealed key drivers of renal function (e.g., eGFR) and the long-term effects of modifiable interventions[21]. Longitudinal targeted maximum likelihood estimation further quantified the attributable mortality of time-varying AKI severity, highlighting the importance of temporal dynamics for ICU kidney protection strategies[22].

In parallel, a large body of work has evaluated feature selection and predictive modeling for CKD and related outcomes. Studies have explored correlation- or sparsity-based selection (e.g., LASSO) and compared classical ML/DL classifiers and ensembles on benchmark datasets[23–25]. Automated model selection and multimodal learning have also been investigated for cardiovascular risk stratification, with interpretability tools (e.g., SHAP) used to identify prognostic factors, while time-censoring and survival bias remain challenges for clinical translation[26–28]. Moreover, causal machine learning frameworks have been applied to estimate heterogeneous treatment effects and quantify drug-related risks, including comparisons between causal forests and penalized regression[29], GAN-based causal inference for treatment effects[30], and broader causal reasoning frameworks for pharmacovigilance[31].

However, in early CKD risk assessment, clinical variables often exhibit substantial redundancy and complex dependencies. Traditional feature selection methods based on mutual information, information gain, or linear correlation can rank importance but cannot distinguish redundant, unique, and synergistic contributions, limiting interpretability and generalizability. To address this gap, SURD decomposes causal information into redundant, unique, and synergistic components, enabling more principled feature selection and interpretable modeling in high-dimensional clinical data. Applying SURD to multimodal CKD variables can identify independently informative indicators and synergistic feature sets that only become discriminative jointly, providing a solid foundation for the modeling tasks in this study.

Experimental study

The experimental process described in this study is shown in Fig. 1. The dataset used in this study comes from the CKD dataset in the UCI repository[32]. The overall experimental process begins with data preprocessing, then utilizes the SURD method to classify the features, and applies different categories to ten different ML techniques. Finally, not only are multiple evaluation indicators used to evaluate the model's performance thoroughly, but, the SHAP visualization method is also employed to explain the essential features.

Datasets

This study utilized the CKD dataset from the UCI database, which is currently employed by many researchers[33]. The dataset has a total of 25 features, of which 14 are nominal and 11 are numerical. More information on this dataset can be found in Table 1. There were a total of 400 participants, of whom 250 were diagnosed with CKD and 150 were healthy. Moreover, the dataset is unbalanced, with 62.5% of cases being CKD (250) and 37.5% being non-CKD (150).

Data processing

Due to the significant amount of missing values in the dataset, as shown in the table, traditional methods such as the mean, median, mode, logistic regression, and KNN imputation may exacerbate issues with missing data issues. Given the substantial number of missing values, it is not feasible to simply delete rows or columns to handle missing data, primarily because the data includes both numerical and categorical types. Two methods can be employed for filling in the missing values. This study first cleans the outliers in the categorical variables. To ensure the objectivity of model evaluation, the dataset is divided in an 8:2 ratio, using stratified sampling to maintain consistency in class distribution. For numerical features, the Multiple Imputation by Chained Equations (MICE) strategy is adopted, which fits the IterativeImputer on the training set and is then applied to both the training and testing sets. For categorical features, the mode imputation (SimpleImputer) is applied

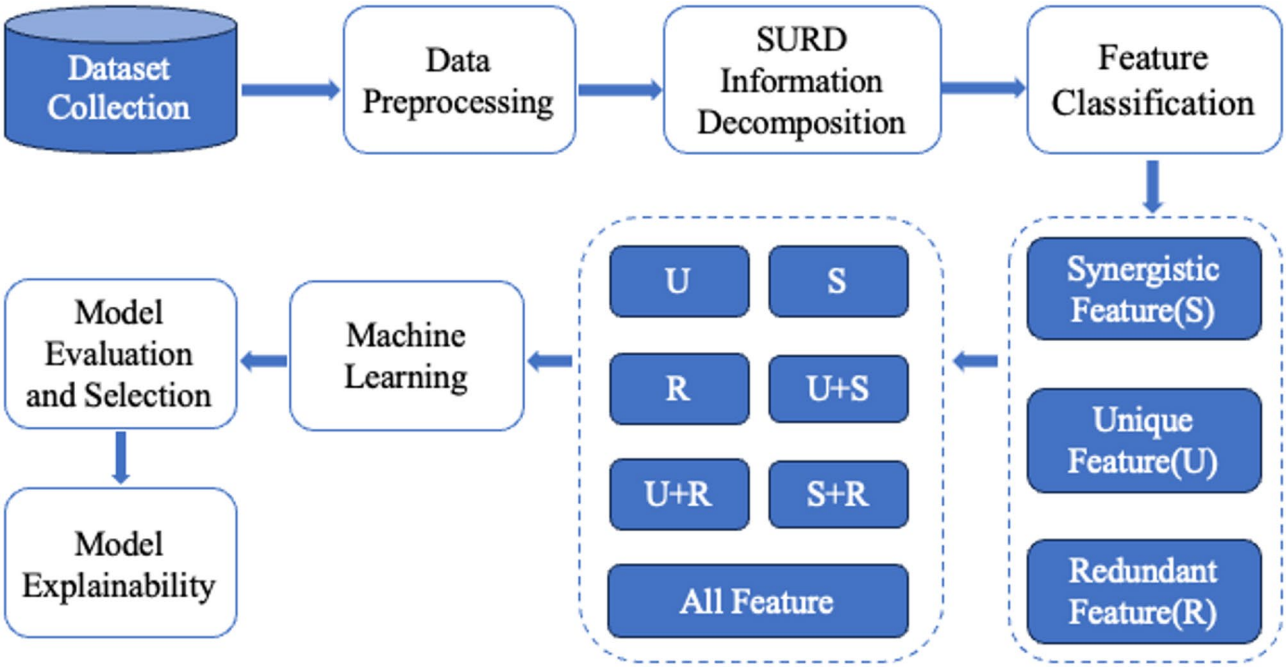

Fig. 1. Flowchart of research methods.

Numerical Table				
No	Attribute	Description	Unit	Missing values
1	Age	Age	Years	9
2	bp	Blood Pressure	mm/Hg	12
10	bgr	Blood Glucose Random	mg/dL	44
11	bu	Blood Urea	mg/dL	19
12	sc	Serum Creatinine	mg/dL	17
13	sod	Sodium Level	mEq/L	87
14	pot	Potassium Level	mEq/L	88
15	hemo	Hemoglobin	g/dL	52
16	pcv	Packed Cell Volume	0–2 integer	70
17	wc	White Blood Cell Count	cells/mm^3	105
18	rc	Red Blood Cell Count	million/mm^3	130
Nominal Table				
No	Attribute	Description	Values/Range	Missing Values
3	sg	Specific Gravity	1.005–1.025	47
4	al	Albumin Level	0–5 ordinal scale	46
5	su	Sugar Level	0–5 ordinal scale	49
6	rbc	Red Blood Cell Morphology	Normal/Abnormal	152
7	pc	Pus Cell Detection	Normal/Abnormal	65
8	pcc	Pus Cell Clumps	Present/Not present	40
9	ba	Bacterial Presence	Present/Not present	4
19	htn	Hypertension Diagnosis	Yes/No	2
20	dm	Diabetes Mellitus Diagnosis	Yes/No	2
21	cad	Coronary Artery Disease	Yes/No	2
22	appet	Appetite Status	Good/Poor	1
23	pe	Pedal Edema	Yes/No	1
24	ane	Anemia Diagnosis	Yes/No	1
25	class	Class	Ckd/Notckd	0

Table 1. UCI ML Repository CKD dataset description with the number of missing values for each attribute.

to the training set, and after fitting, it is utilized on both datasets for the imputed categorical variables, which are then converted into numerical vectors using One-Hot Encoding. The encoder is fitted on the training set and similarly applied to the testing set to ensure consistency in dimensionality. Although the class distribution between CKD and non-CKD samples is moderately imbalanced, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training set to mitigate potential bias toward the majority class and to enhance the robustness of minority-class recognition[34]. All pre-processing operations were strictly fitted on the training set and subsequently applied to the testing set to avoid data leakage, thereby ensuring the reliability of subsequent model training and evaluation.

Data visualization

Before conducting causal inference, a series of preprocessing and visualization steps were applied to the UCI-CKD dataset to explore its structural characteristics. To facilitate effective visualization and reduce analytical complexity, the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm was combined with Principal Component Analysis (PCA) for dimensionality reduction. Based on the reduced feature space, multiple visualization techniques were subsequently employed to examine feature relationships, distributional properties, and target variable characteristics.

Correlation heatmaps were used to visualize the pairwise relationships among features, providing an intuitive overview of both positive and negative correlations. The embedded histograms further complemented the heatmaps by illustrating the distribution of correlation strengths. Through this visualization, highly correlated feature groups and relatively independent variables could be readily identified, offering valuable guidance for subsequent feature selection and for mitigating potential multicollinearity issues.

Violin plots were employed to examine the distributional characteristics of individual features after standardization. These plots illustrate the shape, central tendency, and dispersion of feature distributions, revealing noticeable variability across features despite normalization. Such distributional heterogeneity provides an intuitive understanding of the underlying data structure and informs subsequent modeling decisions.

To investigate the distribution of the target variable, a combination of histograms and kernel density estimates was utilized. The histogram depicts the frequency of samples across outcome categories, while the density curve offers a smoother representation of the overall distribution pattern. As illustrated in subgraph c of Fig. 2, the UCI-CKD dataset exhibits a moderately imbalanced class distribution between CKD and non-CKD samples. This observable asymmetry reflects an inherent structural characteristic of the dataset, which is considered in subsequent model training and evaluation.

In summary, the integrated use of multiple visualization techniques provides a comprehensive overview of feature correlations, distributional differences, and class structure within the dataset. These observations highlight potential multicollinearity, heterogeneity in feature distributions, and class asymmetry, underscoring the importance of careful feature selection and robust modeling strategies in subsequent analyses.

Causal inference and information decomposition

Theoretical foundation: mutual information

The Synergistic-Unique-Redundant Decomposition (SURD) framework is rooted in information theory, utilizing Mutual Information (MI) as a non-parametric measure to quantify dependencies within the joint probability space of clinical features $\boldsymbol{Q}$ and the target diagnostic outcome $\boldsymbol{Q}_j$ (CKD status). The MI is defined as:

$$I\left(Q_j;Q\right)=\sum_{q_j,q}p\left(q_j,q\right)\log_2\left(\frac{p\left(q_j,q\right)}{p\left(q_j\right)p(q)}\right) \tag{1}$$

where $p\left(q_j,\boldsymbol{q}\right)$ is the joint probability distribution, and $p\left(q_j\right)p\left(\boldsymbol{q}\right)$ are the marginal distributions.

Implementation details and data discretization

The study was conducted on the original dataset. To ensure the reproducibility of the MI estimation and subsequent decomposition, we employed an equal-frequency binning strategy for data discretization. Each continuous biomarker dimension was divided into 50 non-overlapping bins, a parameter selection based on prior stability sensitivity analysis which demonstrates that 50 bins achieve a robust balance between computational efficiency and the preservation of system dynamics[15]. In this context, the clinical dataset is interpreted as a stationary realization of a multivariate probability space. From the perspective of Partial Information Decomposition(PID), the informational interaction—specifically how diagnostic information is shared or uniquely contributed—is an intrinsic property of the joint distribution $p\left(q_j,\boldsymbol{q}\right)$ and is not strictly constrained by temporal evolution.This discretization transforms continuous observations into a finite state space,enabling the precise calculation of joint and marginal probabilities.

Specific mutual information decomposition

The core of the SURD implementation lies in the decomposition of specific mutual information $\tilde{i}\left(q_j;\boldsymbol{Q}\right)$,which quantifies the local dependency at each discrete state q_j of the target variable:

$$\tilde{i}\left(q_j;\boldsymbol{Q}\right)=\sum_{\boldsymbol{q}}\frac{p\left(q_j,\boldsymbol{q}\right)}{p\left(q_j\right)}\log_2\left(\frac{p\left(q_j,\boldsymbol{q}\right)}{p\left(q_j\right)p\left(\boldsymbol{q}\right)}\right) \tag{2}$$

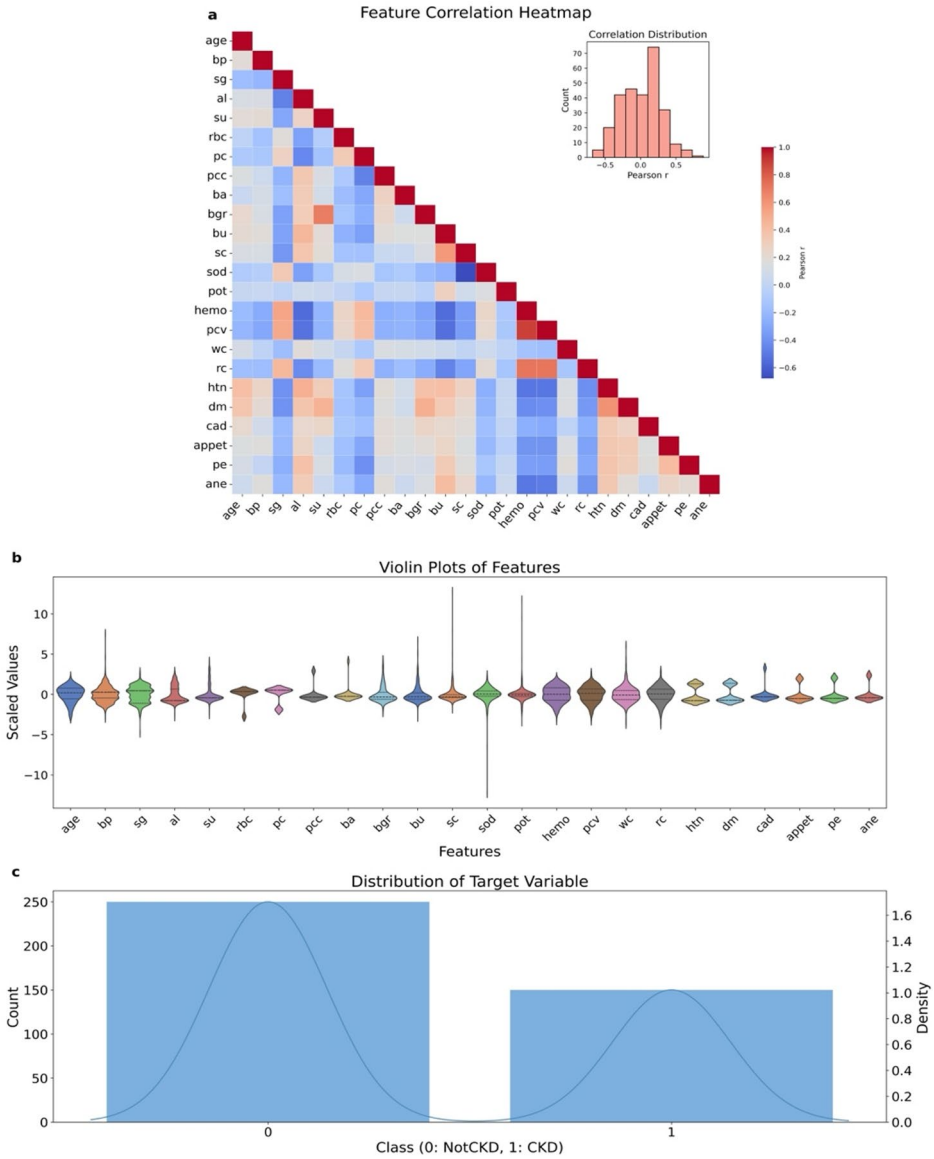

Fig. 2. Dataset visualization. (**a**): Correlation heatmap;(**b**): Violin plot; (**c**): Distribution of Target Variable.

For a given state q_j,the framework traverses all non-empty subsets of the feature vector $\boldsymbol{Q}$.The informational contributions of these subsets are organized in ascending order to assign the synergistic, unique, and redundant components:

Redundant information($\Delta\tilde{\imath}_i^R$) : The minimum amount of information provided by multiple subsets reflects overlapping predictive capabilities.

Unique information($\Delta\tilde{\imath}_i^U$) : The amount of intermediate information provided independently by a single subset reflects its unique contribution.

Synergistic information($\Delta\tilde{\imath}_i^S$) : The maximum amount of information that requires the cooperation of multiple subsets to provide, reflecting non-linear interaction effects.

The above process is conducted independently for all possible states q_j^+ of the target variable. Finally, by calculating the expected values of each state component, the forms of mutual information decomposition (3) (4) (5) are obtained.

$$\Delta\tilde{\imath}_{i\to j}^R = \sum_{q_j^+} p\left(q_j^+\right) \Delta\tilde{\imath}_i^R\left(q_j^+\right) \tag{3}$$

$$\Delta\tilde{\imath}_{i\to j}^U = \sum_{q_j^+} p\left(q_j^+\right) \Delta\tilde{\imath}_i^U\left(q_j^+\right) \tag{4}$$

$$\Delta \tilde{\imath}^{S}_{i \to j} = \sum_{q_j^+} p\left(q_j^+\right) \Delta \tilde{\imath}_i^{S}\left(q_j^+\right) \tag{5}$$

Global Information Integration

Ultimately, the global decomposition metrics for the CKD model are derived by calculating the expected values of the local components across the marginal distribution of the diagnostic outcomes:

$$\Delta I_{i \to j} = \sum_{q_j} p\left(q_j\right) \Delta \tilde{\imath}_i\left(q_j\right) \tag{6}$$

Through this mathematical path of hierarchical decomposition and integration, this study not only achieves the identification of key predictive factors but also unveils the intrinsic informational structure of disease mechanisms from an information-theoretic perspective. This approach transforms local state-dependencies into a global feature profile, providing a rigorous and reproducible theoretical foundation for feature selection and mechanistic interpretation in complex diseases.

Model training and testing

The feature subsets were identified using the training set through SURD-based causal information decomposition and then evaluated on the independent test set across ten classifiers[35]. Given the high dimensionality and non-linear nature of clinical chronic kidney disease (CKD) data, we selected models across a spectrum of computational paradigms, focusing on the trade-off between interpretability and predictive depth.

Standard learners, including Decision Trees (DT)[36], Naive Bayes (NB)[37], K-Nearest Neighbors (KNN)[38], and Support Vector Machines (SVM)[39], were utilized to establish a performance baseline. These algorithms are prioritized in clinical informatics for their transparency, which is essential for verifying that the model's decision logic aligns with established physiological markers of renal decline. To address potential noise and complex feature interactions within the observational trial data, we leveraged ensemble architectures—Random Forest (RF)[40], XGBoost[41], and AdaBoost[42]. These models provide a robust assessment of feature stability through bagging and boosting mechanisms, which mitigate the risk of overfitting inherent in small-to-moderate clinical datasets.

While the UCI-CKD dataset is primarily static and non-temporal, we intentionally incorporated Multilayer Perceptron (MLP)[43], Recurrent Neural Networks (RNN)[44], and Long Short-Term Memory (LSTM)[45] as high-capacity benchmarks. The inclusion of sequential architectures (RNN and LSTM) serves as a rigorous stress test for our proposed framework; we aimed to empirically determine whether the increased architectural depth of deep learning offers any incremental advantage over more efficient, traditional models in a static tabular context. This comprehensive benchmarking ensures that the selected feature set maintains strong discriminatory power regardless of the underlying algorithmic complexity.

Furthermore, We introduced LASSO[46] and the PC algorithm[47] as baselines to isolate the impact of the SURD framework. While LASSO represents a purely statistical approach to feature sparsity, the PC algorithm serves as a representative of classical constraint-based causal discovery. This comparison allows us to verify whether the predictive gains stem from the causal information-theoretic decomposition itself, rather than from a specific model architecture or standard statistical regularization.

Model evaluation metrics

Model performance was validated through fivefold cross-validation to ensure the generalizability of the results across diverse patient subsets. In accordance with established clinical reporting standards, we quantified model efficacy using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic curve (AUC)[48–50].

Model hyperparameters

In this study, grid search combined with fivefold stratified cross-validation was employed to optimize the hyperparameters of the classifiers. To address potential class imbalances and ensure model robustness, the Area Under the ROC Curve was utilized as the primary optimization criterion to select the best-performing configurations. Both the predefined search ranges and the resulting optimal hyperparameters for the 10 models used in this study are detailed in Supplementary Table S1.

Experimental setup

All studies were conducted on Python 3.12 on an Apple M4-chipped MacBook Pro with 16 GB of RAM, and the operating system was macOS Sonoma 15.3.

Results and discussions

Table 2 and Figure 3 show the distributions of three information components—synergistic, unique, and redundant—computed using the SURD information decomposition method across three corresponding feature groups. The box plots with superimposed scatter points illustrate the value ranges and dispersion of each information component. Non-parametric Mann–Whitney U tests confirm that the differences among the three feature groups are statistically significant across all information components ($p < 0.05$).

Based on the SURD information decomposition method, this study divides the original features into synergistic, unique and redundant. In order to evaluate the impact of different classification feature combinations

Feature	Redundant	Unique	Synergistic	Category
sg	0.096624	2.78E-01	0.074243	S
bgr	0.080132	6.10E-02	0.064159	S
dm	0.086167	8.09E-02	0.06366	S
htn	0.090538	1.03E-01	0.061582	S
hemo	0.096624	1.98E-01	0.056311	S
pcv	0.069621	3.74E-02	0.048052	R
pc	0.057078	1.16E-02	0.046689	R
al	0.056653	1.10E-02	0.044765	R
appet	0.052669	7.65E-03	0.044497	R
rc	0.052704	1.40E-02	0.043927	R
pe	0.048012	5.18E-03	0.042774	R
bp	0.045126	4.04E-03	0.042257	R
su	0.055258	9.51E-03	0.042146	R
bu	0.048572	5.26E-03	0.038657	R
ane	0.035785	2.12E-03	0.035203	U
rbc	0.026533	8.56E-04	0.032997	U
pcc	0.023163	5.44E-04	0.025322	U
cad	0.018074	2.69E-04	0.017219	U
age	0.013432	1.08E-04	0.016297	U
ba	0.010945	4.39E-05	0.013214	U
pot	0.006127	5.67E-06	0.012646	U
wc	0.003262	9.51E-19	0.006618	U
sc	0.001725	1.83E-17	0.003116	U
sod	0.001278	2.06E-17	0.00251	U

Table 2. Contribution values of classification features based on SURD. "S": Synergistic. "R": Redundant. "U": Unique.

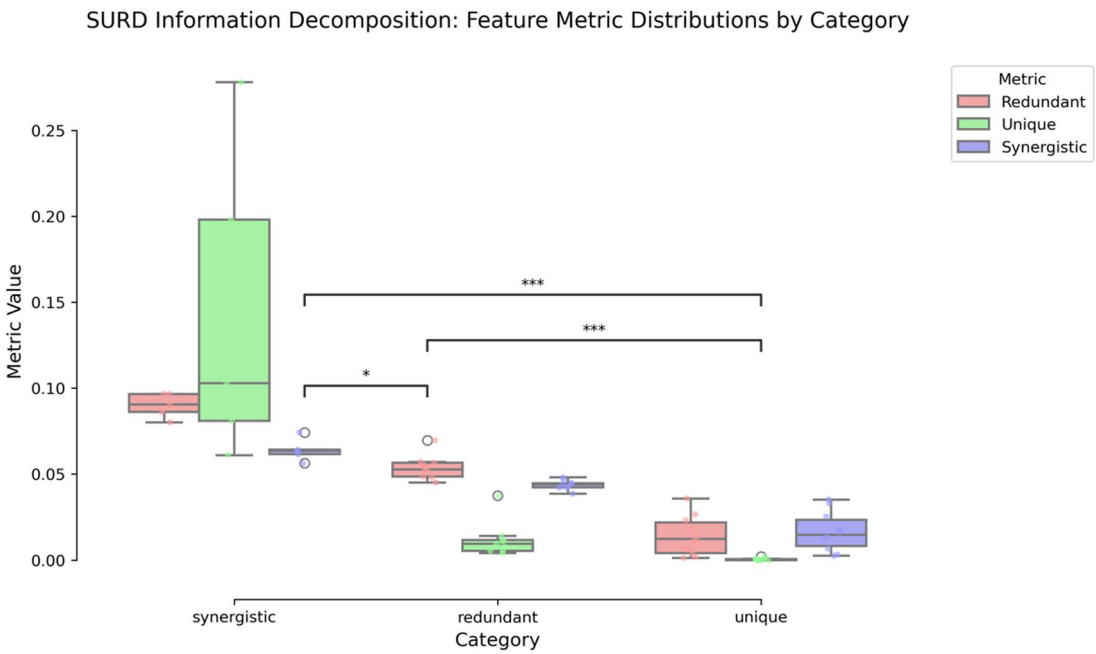

Fig. 3. Distribution of SURD information decomposition indicators (redundant, unique, and collaborative information) in different feature categories (synergistic, redundant, and unique). The box plot shows the numerical distribution and dispersion of the three information indicators in each category, and the scatter plot overlay shows the specific values of a single feature. The significant asterisk is based on the Mann–Whitney U test, indicating that the difference in indicators between categories is statistically significant.

Classifier	Accuracy	Precision	Recall	F1 score	AUC
AdaBoost	0.912	0.705	0.857	0.774	0.890
DT	0.987	0.965	1.000	0.982	0.990
KNN	0.925	0.823	1.000	0.903	0.942
LSTM	0.967	0.955	0.958	0.955	0.995
MLP	0.987	0.988	0.987	0.987	0.999
NB	0.962	0.963	0.928	0.945	0.997
RF	1.000	1.000	1.000	1.000	1.000
RNN	0.975	0.976	0.975	0.975	0.997
SVM	0.987	0.965	1.000	0.982	0.997
XGBoost	0.975	0.933	1.000	0.965	0.991

Table 3. Performance comparison using all features.

Classifier	Accuracy	Precision	Recall	F1 score	AUC
AdaBoost	0.975	0.933	1.000	0.965	1.000
DT	1.000	1.000	1.000	1.000	1.000
KNN	0.962	0.903	1.000	0.949	0.989
LSTM	0.972	0.960	0.968	0.963	0.995
MLP	0.970	0.972	0.970	0.970	0.996
NB	0.987	0.965	1.000	0.982	0.960
RF	1.000	1.000	1.000	1.000	1.000
RNN	0.987	0.987	0.987	0.987	0.993
SVM	0.975	0.933	1.000	0.965	1.000
XGBoost	0.987	0.965	1.000	0.982	1.000

Table 4. Performance comparison using U + S feature combination.

on model performance, the study constructed synergistic, unique, redundant, unique and synergistic (U + S), unique and redundant (U + R), Synergistic + Redundant(S + R) models, and cross-validated them on ten classification algorithms. The output of each model is estimated using various evaluation indicators (such as accuracy, precision, recall, F-measure and ROC-AUC).

Models constructed using unique features alone generally exhibited lower and less stable predictive performance across classifiers. Detailed results are provided in Table S2. Synergistic feature–based models achieved strong performance in certain classifiers, although their results were less consistent across models, as summarized in Table S3. Models relying solely on redundant features demonstrated high accuracy in some cases, with overall performance variability shown in Table S4.

To further validate the practical effectiveness of SURD-based feature selection, the performance of models using the full feature set was compared with that of models constructed using the U + S feature combination across ten classifiers. Detailed results are reported in Table 3 and .

Table 4. Overall, the U + S feature combination achieved superior performance in most models, particularly with respect to F1 score and AUC. Performance gains were most pronounced in DT, RF, XGBoost, SVM, MLP, and NB models.

ROC curves and corresponding AUC values further supported the classification performance of the evaluated models, as shown in Fig. 4. All models achieved high AUC values, with 95% confidence intervals reported to quantify statistical uncertainty. AdaBoost, DT, SVM, RF, and XGBoost achieved an AUC of 1.000 (95% CI 1.000–1.000), while other models also demonstrated strong and stable performance, including KNN (AUC: 0.989, 95% CI 0.963–1.000) and NB (AUC: 0.960, 95% CI 0.917–0.991).

Beyond discriminative performance, clinical utility was evaluated using Decision Curve Analysis (DCA), as shown in Fig. 5. All models exhibited consistently positive net benefit across a wide range of threshold probabilities, yielding higher net benefit than both treating all individuals and treating none across most threshold probabilities. Notably, the RF, XGBoost, and SVM models achieved the highest net benefit and maintained stable performance even at higher risk thresholds.

Overall, these results demonstrate that the U + S feature combination derived from the SURD framework consistently improves classification performance across multiple models while maintaining robust discriminative ability and clinical utility.

Comparative analysis of feature selection frameworks

To validate SURD's advantages, we compared its performance against LASSO and the PC algorithm (detailed results in Supplementary Tables S5 and S6; visualizations in Figures S1 and S2).

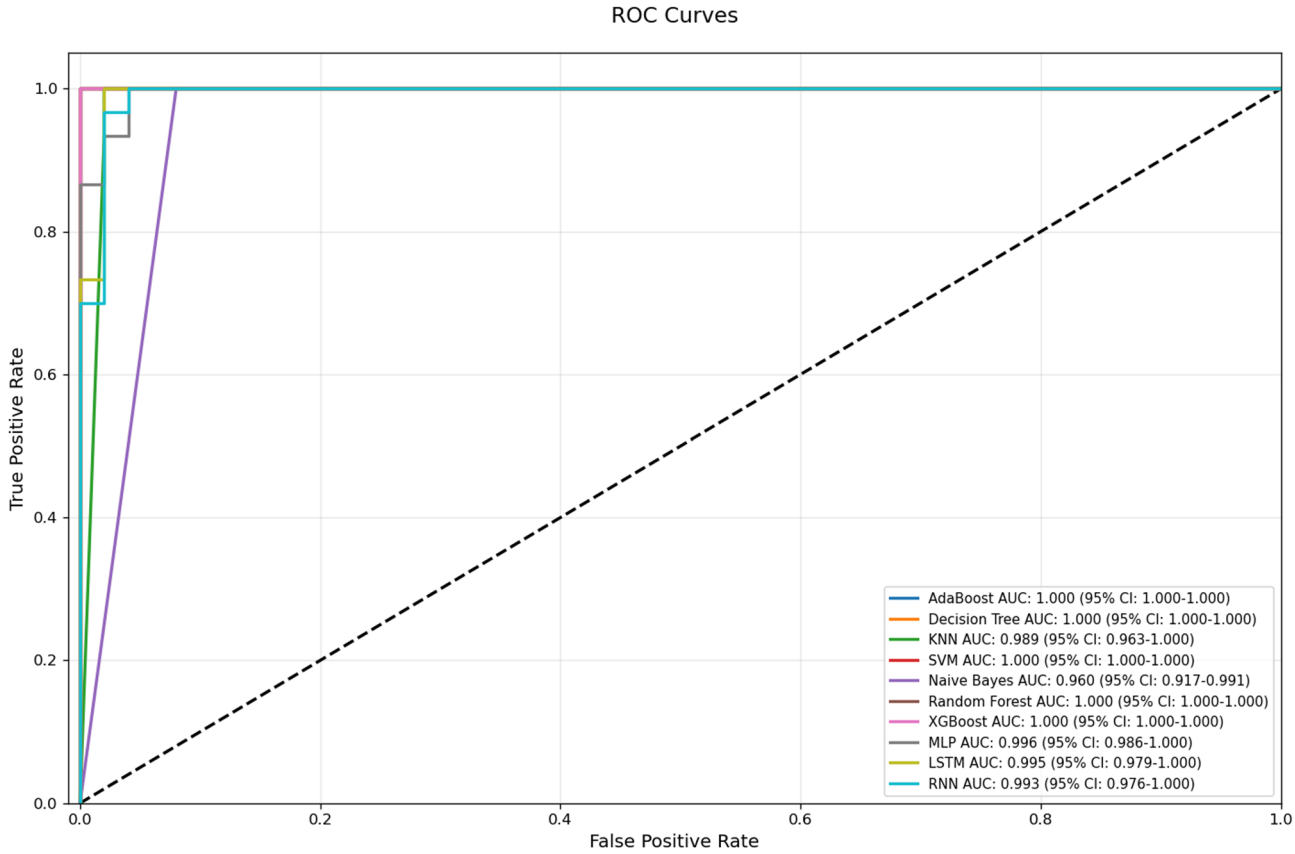

Fig. 4. ROC curves of ten machine learning models using U + S feature combinations.

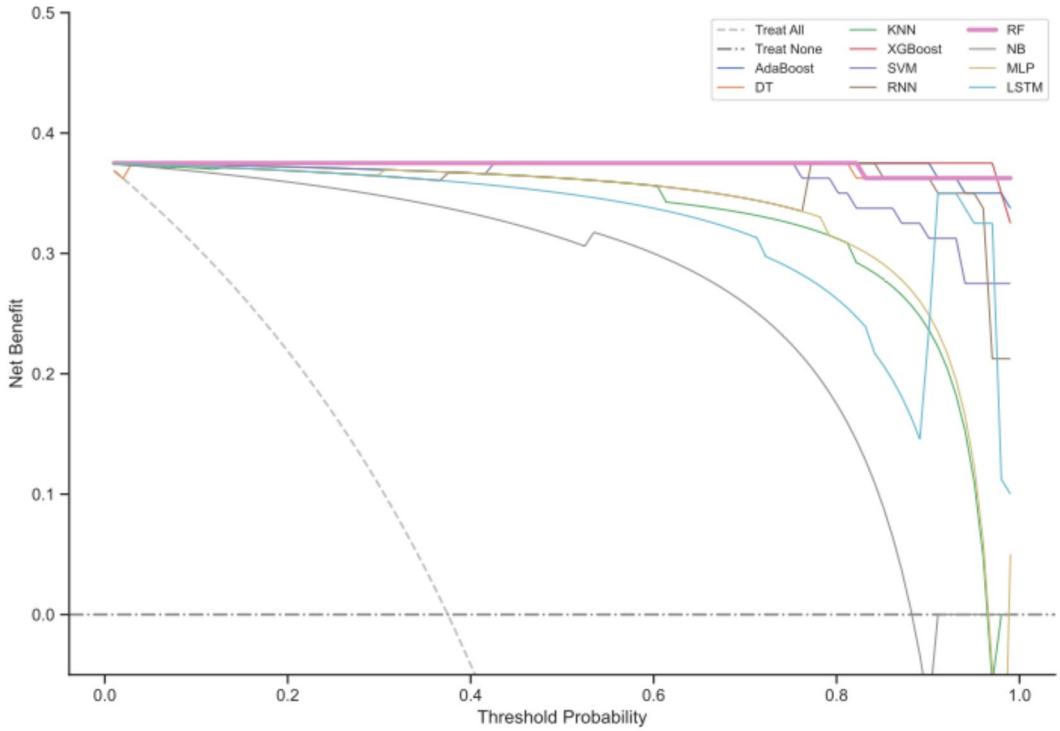

Fig. 5. Decision Curve Analysis Comparing Net Benefit Across Different Models.

The results indicate that while all methods achieved high accuracy, SURD demonstrated superior algorithmic robustness. Specifically, under the PC-derived feature set, the NB classifier's accuracy dropped to 0.907, and under LASSO, its AUC was 0.963. In contrast, the SURD-based U + S combination maintained near-optimal performance across all models, including NB (AUC = 0.997).

This discrepancy suggests that traditional methods like PC and LASSO may fail to capture the synergistic dependencies essential for stable prediction. The observed performance saturation (AUC > 0.99) across paradigms confirms that the predictive ceiling is driven by causal feature quality rather than model complexity, justifying our focus on information decomposition over black-box architectures like GNNs.

To further ensure a fair comparison and evaluate whether model performance was biased by complexity, we analyzed the learning curves for the classifiers Figure S3. Taking the Random Forest model as a representative case, the training score and cross-validation score converge effectively as the number of training examples increases beyond 100. This convergence indicates that the high performance is a result of effective learning of the underlying data patterns rather than overfitting or unfair complexity advantages.

External validation

To evaluate the robust generalizability of our model, we conducted an independent external validation using the MIMIC-IV database (n = 27,834)[51]. The author, Ningning Xue, has completed the Collaborative Institutional Training Initiative (CITI Program) and received the corresponding certification (record number:14856183). Despite the inherent differences in baseline characteristics between the two datasets, our model demonstrated robust generalizability. The externally validated ROC curve for the Random Forest model yielded an AUC of 0.990 (95% CI 0.989–0.991), as illustrated in Figure S4, which significantly outperformed the Decision Tree baseline (AUC: 0.914, 95% CI0.912–0.916).

Model explanation and feature importance analysis

This study employs SHAP (Shapley Additive exPlanations) to characterize the contribution of individual features to model predictions. SHAP values quantify the marginal contribution of each feature to the prediction outcome and provide a model-specific view of feature relevance. In this study, SHAP was used to rank feature importance for the RF model, thereby highlighting variables that exert strong influence on predictive outputs. To complement this model-based interpretation, SHAP results were examined alongside SURD-based information decomposition to further explore the structural roles of features.

The SHAP analysis identified hemo, sg, sc, htn, and dm as the most influential features in CKD prediction. These variables consistently showed substantial contributions to model outputs. SHAP values were computed for both the RF and DT models; due to space limitations, only the RF results are presented in Fig. 6, while the corresponding DT results are provided in Figure S5.

To quantitatively assess the relationship between SHAP-based feature attribution and information-theoretic relevance, we computed Spearman rank correlations between SHAP global importance scores and SURD-derived relevance measures (U + S). A moderate correlation was observed (Spearman's $\rho = 0.50$, $p = 0.0577$), indicating partial concordance between the two approaches. This result suggests that SHAP and SURD capture overlapping but distinct aspects of feature importance.

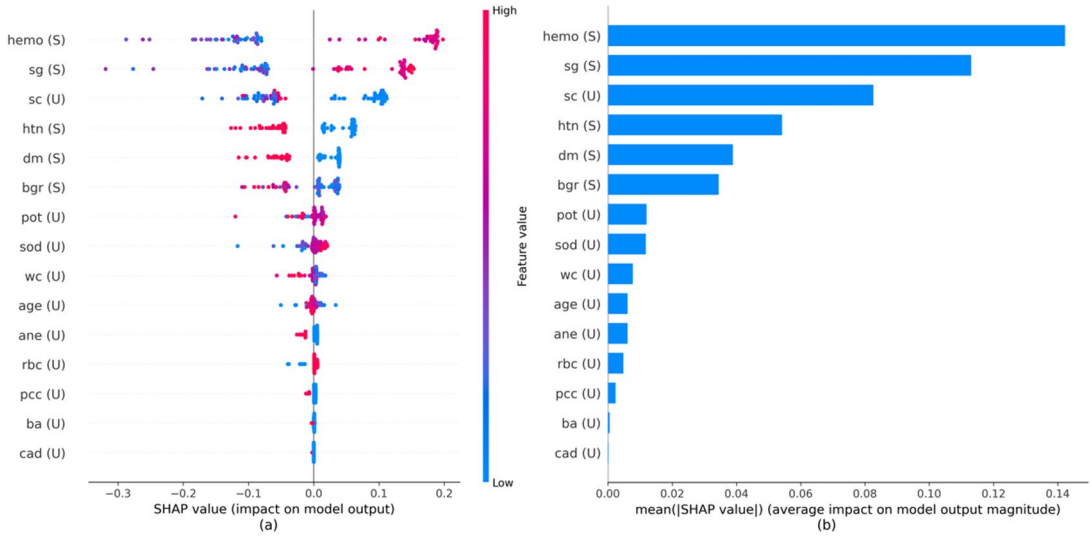

Fig. 6. Visual explanation of predictive CKD based on the Random Forest prediction model. (**a**)Each dot represents a sample, and the color of the dot indicates the high or low feature value of the sample (red = high value, blue = low value); the horizontal position of the dot indicates the direction and intensity of the impact of the feature value on the model output. (**b**)Bar chart of the average feature importance of each feature SHAP value amplitude.

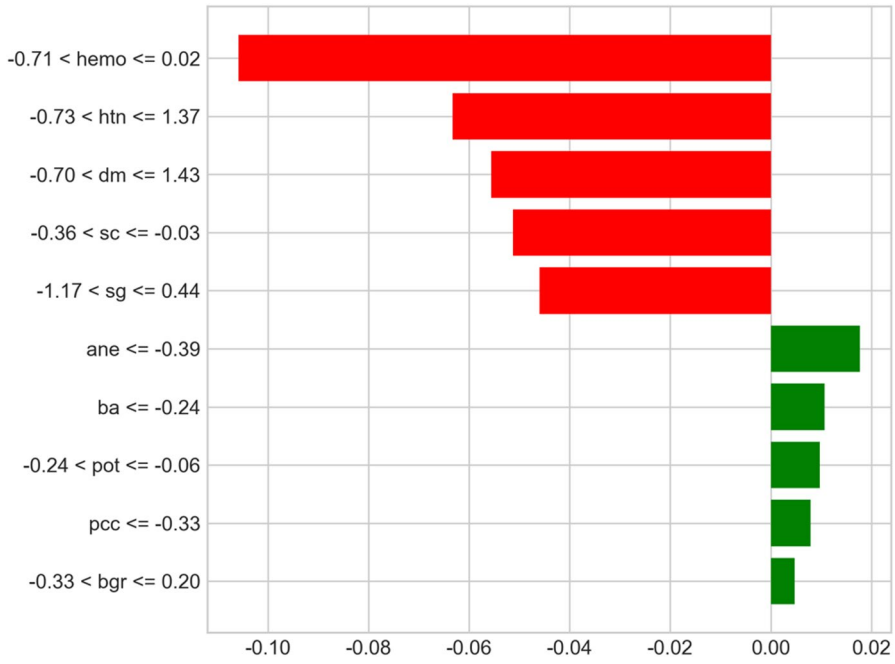

Fig. 7. Local interpretability analysis using LIME. This plot illustrates the feature contributions for a representative instance correctly classified as 'Normal' (True Label: 0). Red bars (left) represent features that support the 'Normal' prediction by negating the probability of 'CKD', while green bars (right) indicate features with a marginal tendency toward the 'CKD' class in this specific case.

Proposer	Features	Method	Evaluation indicators
Ashwan et al.[5]	7	RFE and DNN	Accuracy:98.33%
K.M. et al.[2]	6	Filter-based、Wrapper methods and RF	Accuracy:0.98%,Precision:0.98%,Recall:0.99% F1-Score:0.98%
Bishnu et al.[24]	14	Hybrid Model	Accuracy:94.99%,Precision:95.21%,F1-Score:95.32%,Recall:95.11%,AUC-ROC:95.56%
Shiddarth et al.[23]	14	RF、XGBoost	Accuracy:0.97%,Precision:0.98%,Recall:0.97%, F1-Score:0.97%
This study	16	SURD and DT、RF	Accuracy:100.00%,Precision:100.00%,F1-Score:100.00%,Recall:100.00%,AUC-ROC:100.00%

Table 5. Comparison of UCI-CKD dataset.

From the SURD decomposition, most features with high SHAP importance (with the exception of sc) were primarily characterized by synergistic information, indicating that their predictive contributions arise mainly through interactions with other variables. In contrast, sc exhibited a dominant unique information component, consistent with its role as an independent indicator of renal function in clinical practice.

Overall, SHAP highlights features that strongly influence model predictions, whereas SURD provides complementary insight into whether such influence is driven by unique or interaction-based information structures. The combined analysis therefore supports a multi-level interpretation of feature relevance rather than a direct equivalence between the two methods.

To further support interpretability at the individual level, LIME was applied to generate local explanations for representative cases, as illustrated in Fig. 7. Features such as hemo, htn, and dm were identified as major contributors, consistent with the dominant patterns observed in the SHAP analysis.

Comparative analysis

To comprehensively assess the latest progress in the field, this section provides a comprehensive and systematic comparative analysis. Table 5 summarises representative studies recently related to our work, with each study using the CKD dataset from the UCI ML database and employing different methods to predict CKD. The table highlights significant differences among the studies in feature selection methods, modelling techniques, and evaluation metrics. For instance, K.M. et al.[2] combined filter and wrapper methods to select 6 features, achieving optimal results with an RF model (Accuracy: 0.98%, Recall: 0.99%).Meanwhile, Ashwan et al.[5]selected 7 features, combining RFE and DNN models to achieve an accuracy rate of 98.33%.

Furthermore, Shiddarth et al.[23] and Bishnu et al.[24] both used 14 features, constructing hybrid models and ensemble learning models (RF, XGBoost), and achieving excellent predictive performance. However, most of

these studies did not systematically address the issue of class imbalance and primarily adopted traditional feature selection methods.

In contrast, the research presented in this paper optimizes and improves upon several aspects. Firstly, during the data preprocessing stage, MICE was introduced to handle missing values, and the SMOTE technique was effectively employed to alleviate the class imbalance problem. Secondly, in feature selection, the SURD method was used for an in-depth analysis of variable information, ultimately selecting 16 representative features. Additionally, during the modeling phase, multiple models, such as DT and RF, were employed for cross-validation, further enhancing the stability and accuracy of the predictions.

In terms of evaluation metrics, this study achieved 100% across various dimensions, including accuracy, precision, recall, F1 score, and AUC-ROC value, clearly surpassing the highest levels in current literature. This indicates that the methods employed in this study not only possess stronger causal decomposition capabilities but also exhibit outstanding generalization ability in models, demonstrating higher clinical application potential.

Conclusion

This study proposes an information-theoretic framework that combines SURD-based information decomposition with machine learning for chronic kidney disease risk assessment. By decomposing clinical variables into unique, synergistic, and redundant components, this method provides a structured perspective on feature causality, surpassing traditional correlation-based feature selection strategies and offering deeper insights into the informational roles of clinical variables.

Using the UCI-CKD dataset as a benchmark, the results indicate that SURD-informed feature groupings support stable discrimination across multiple representative machine learning models. Rather than emphasizing absolute performance values, these findings highlight the consistency and interpretability of the feature space induced by information decomposition across different model families.

To examine robustness beyond a single cohort, the framework was further evaluated on the MIMIC-IV database ($n = 27{,}834$), where it achieved a stable AUC of 0.990 (95% CI 0.989–0.991). This result offers supporting evidence that the feature hierarchies identified by SURD remain informative across datasets with differing institutional characteristics, while not constituting definitive clinical validation.

With respect to interpretability, the observed Spearman rank correlation between SHAP-based feature attribution and SURD-derived relevance ($\rho = 0.50$, $p = 0.0577$) indicates partial concordance. This suggests that model-based importance and information-theoretic structure capture complementary, rather than redundant, aspects of feature relevance.

This study constructs an analysis framework based on the combination of SURD information decomposition and a prediction model, which demonstrating methodological value for structured feature analysis and interpretable risk modeling. Future research can further expand the following aspects.

First, time series data can be introduced within the existing framework, and by combining dynamic Bayesian networks or time-cause modeling methods based on deep learning, the temporal change patterns and potential causal paths between variables can be thoroughly explored. This will help to more comprehensively understand the interactions and masking trends of key clinical indicators at different time points, thereby enhancing the model's perception and interpretation of dynamic features.

Secondly, it is recommended to apply this analysis framework to larger-scale and more diverse sample populations, especially in combination with real-world clinical electronic health record (EHR) data, further to verify its applicability and robustness in different clinical scenarios. This can not only enhance the practical application capabilities of the model but also help promote the widespread promotion of information decomposition methods in the field of medical health.

Overall, this study demonstrates the methodological potential of combining information decomposition with machine learning for structured feature analysis and interpretable risk modeling.

Data availability

The datasets analysed during the current study are available in the UCI Machine Learning Repository: [https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease. MIMIC-IV datasets: [https://physionet.org/content/mimiciv/3.1/]. The main code mentioned in the article can be accessed at https://github.com/xueningning-052/CKD.

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References

1. Hoerger, T. J. et al. The future burden of CKD in the united states: A simulation model for the CDC CKD initiative. *Am. J. Kidney Dis.* **65**, 403–411 (2015).
2. Tawsik Jawad, K. M., Verma, A., Amsaad, F. & Ashraf, L. A study on the application of explainable AI on ensemble models for predictive analysis of chronic kidney disease. *IEEE Access* **13**, 23312–23330 (2025).
3. S, S., A, R. K. & K, L. A comprehensive analysis of machine learning algorithms in diagnosis of chronic kidney disease. in *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)* 1–5 (IEEE, Kharagpur, India, 2022). https://doi.org/10.1109/ICCCNT54827.2022.9984275.
4. Romagnani, P. et al. Chronic kidney disease. *Nat. Rev. Dis. Primers* **3**, 17088 (2017).
5. Implementation of clinical diagnosis system for chronic kidney disease using deep learning algorithms. *Int. J. Adv. Technol. Eng. Explor.* **11**, (2024).
6. Richens, J. G., Lee, C. M. & Johri, S. Improving the accuracy of medical diagnosis with causal machine learning. *Nat. Commun.* **11**, 3923 (2020).

7. Cheng, J., Bell, D. A. & Liu, W. Learning belief networks from data: an information theory based approach. in *Proceedings of the sixth international conference on Information and knowledge management* 325–331 (ACM, Las Vegas Nevada USA, 1997). https://doi.org/10.1145/266714.266920.
8. Tsagris, M. Bayesian network learning with the PC algorithm: An improved and correct variation. *Appl. Artif. Intell.* **33**, 101–123 (2019).
9. Tsamardinos, I., Brown, L. E. & Aliferis, C. F. The max-min hill-climbing bayesian network structure learning algorithm. *Mach. Learn.* **65**, 31–78 (2006).
10. Gasse, M., Aussem, A. & Elghazel, H. A hybrid algorithm for bayesian network structure learning with application to multi-label learning. *Expert Syst. Appl.* **41**, 6755–6772 (2014).
11. Granger, C. W. J. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* **37**, 424 (1969).
12. Sugihara, G. et al. Detecting causality in complex ecosystems. *Science* **338**, 496–500 (2012).
13. Schreiber, T. Measuring information transfer. *Phys. Rev. Lett.* **85**, 461–464 (2000).
14. Runge, J. Conditional independence testing based on a nearest-neighbor estimator of conditional mutual information.
15. Martínez-Sánchez, Á. Decomposing causality into its synergistic, unique, and redundant components.
16. Talukder, M. A., Talaat, A. S. & Kazi, M. HXAI-ML: A hybrid explainable artificial intelligence based machine learning model for cardiovascular heart disease detection. *Results Eng.* **25**, 104370 (2025).
17. Talukder, M. A., Talaat, A. S., Kazi, M. & Khraisat, A. XAI-HD: An explainable artificial intelligence framework for heart disease detection. *Artif. Intell. Rev.* **58**, 385 (2025).
18. Zhang, M. et al. Development and validation of a multi-causal investigation and discovery framework for knowledge harmonization (MINDMerge): A case study with acute kidney injury risk factor discovery using electronic medical records. *Int. J. Med. Inf.* **191**, 105588 (2024).
19. Noh, M. J. & Kim, Y. S. Diabetes prediction through linkage of causal discovery and inference model with machine learning models. *Biomedicines* **13**, 124 (2025).
20. Zheng, Q. *et al.* Machine learning algorithms to predict stroke in China based on causal inference of time series analysis. Preprint at https://doi.org/10.48550/arXiv.2503.14512 (2025).
21. Petousis, P., Gordon, D., Nicholas, S. B. & Bui, A. A. T. Testing causal explanations: A case study for understanding the effect of interventions on chronic kidney disease. Preprint at https://doi.org/10.48550/arXiv.2410.12047 (2024).
22. Demares, A. Causal machine learning of the attributable mortality of ICU-acquired acute kidney injury in critically ill patients.
23. Tusar, S. D. Advancing chronic kidney disease prediction through machine learning and deep learning with feature analysis. *Open Access* **13**,.
24. Bishnu Padh Ghosh *et al.* Advancing chronic kidney disease prediction: Comparative analysis of machine learning algorithms and a hybrid model. *J. Comput. Sci. Technol. Stud.* **6**, 15–21 (2024).
25. Chittora, P. et al. Prediction of chronic kidney disease - A machine learning perspective. *IEEE Access* **9**, 17312–17334 (2021).
26. Shishehbori, F. & Awan, Z. Enhancing cardiovascular disease risk prediction with machine learning models.
27. Wang, T.-H., Kao, C.-C. & Chang, T.-H. Ensemble machine learning for predicting 90-day outcomes and analyzing risk factors in acute kidney injury requiring dialysis. *J. Multidiscip. Healthc.* **17**, 1589–1602 (2024).
28. Talukder, M. A. et al. An explainable deep learning framework for trustworthy arrhythmia detection from ECG signals. *Sci. Rep.* **15**, 39496 (2025).
29. Venkatasubramaniam, A. et al. Comparison of causal forest and regression-based approaches to evaluate treatment effect heterogeneity: An application for type 2 diabetes precision medicine. *BMC Med. Inform. Decis. Mak.* **23**, 110 (2023).
30. Kang, M. W. & Kang, Y. Utilizing deep learning-based causal inference to explore vancomycin's impact on continuous kidney replacement therapy necessity in blood culture-positive intensive care unit patients. *Microbiol. Spectr.* **13**, e02662-24 (2025).
31. Zhao, Y. et al. Machine learning in causal inference: Application in pharmacovigilance. *Drug Saf.* **45**, 459–476 (2022).
32. L. Rubini, P. S. 2015 Chronic kidney disease. UCI Machine Learning Repository https://doi.org/10.24432/C5G020
33. Saleh, E. & Bin Abd Kadir, M. F. Prediction of chronic kidney disease using data mining techniques. *SSRN Electron. J.* **1**, 1 (2022).
34. Elreedy, D., Atiya, A. F. & Kamalov, F. A theoretical distribution analysis of synthetic minority oversampling technique (SMOTE) for imbalanced learning. *Mach. Learn.* **113**, 4903–4923 (2024).
35. Silveira, A. C. M. D. et al. Exploring early prediction of chronic kidney disease using machine learning algorithms for small and imbalanced datasets. *Appl. Sci.* **12**, 3673 (2022).
36. Prasad, A. M., Iverson, L. R. & Liaw, A. Newer classification and regression tree techniques: Bagging and random forests for ecological prediction. *Ecosystems* **9**, 181–199 (2006).
37. Zhang, Z. Naïve bayes classification in R.
38. Denoeux, T. A k-nearest neighbor classification rule based on dempster-shafer theory. *IEEE Trans. Syst. Man Cybern.* **25**, 804–813 (1995).
39. Boser, B. E., Guyon, I. M. & Vapnik, V. N. A training algorithm for optimal margin classifiers. in *Proceedings of the fifth annual workshop on Computational learning theory* 144–152 (ACM, Pittsburgh Pennsylvania USA, 1992). https://doi.org/10.1145/130385.130401.
40. Breiman, L. Random forests. *Mach. Learn.* **45**, 5–32 (2001).
41. Chen, T. & Guestrin, C. XGBoost: A scalable tree boosting system. in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 785–794 (ACM, San Francisco California USA, 2016). https://doi.org/10.1145/2939672.2939785.
42. Freund, Y. & Schapire, R. E. A decision-theoretic generalization of on-line learning and an application to boosting. *J. Comput. Syst. Sci.* **55**, 119–139 (1997).
43. Singh, J. & Banerjee, R. A Study on Single and Multi-layer Perceptron Neural Network. in *2019 3rd International Conference on Computing Methodologies and Communication (ICCMC)* 35–40 (IEEE, Erode, India, 2019). https://doi.org/10.1109/ICCMC.2019.8819775.
44. Wang, J. & Chankong, V. Recurrent neural networks for linear programming: Analysis and design principles. *Comput. Oper. Res.* **19**, 297–311 (1992).
45. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9**, 1735–1780 (1997).
46. Roth, V. The generalized LASSO. *IEEE Trans. Neural Netw.* **15**, 16–28 (2004).
47. Le, T. D. et al. A Fast PC algorithm for High Dimensional Causal Discovery with Multi-Core PCs. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **16**, 1483–1495 (2019).
48. Bai, T. et al. Exploration and comparison of the effectiveness of swarm intelligence algorithm in early identification of cardiovascular disease. *Sci. Rep.* **15**, 4647 (2025).
49. Xu, C., Shi, F., Ding, W., Fang, C. & Fang, C. Development and validation of a machine learning model for cardiovascular disease risk prediction in type 2 diabetes patients. *Sci. Rep.* **15**, 32818 (2025).
50. Guan, C. et al. Interpretable machine learning model for new-onset atrial fibrillation prediction in critically ill patients: A multi-center study. *Crit. Care* **28**, 349 (2024).
51. Johnson, A. E. W. et al. MIMIC-IV, a freely accessible electronic health record dataset. *Sci. Data* **10**, 1 (2023).

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Declarations

Competing interests
The authors declare no competing interests.

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